February 2025

The Syllabus for Basic Academic Abilities in the EJU

(To be applied to the questions of the 2026 EJU 1st Session)

〈 Syllabus for Science 〉

[Purpose of the Examination]

The purpose of this examination is to test whether international students have the basic academic ability in science necessary for studying at universities or other higher educational institutions in Japan.

[Classification of the Examination]

The examination consists of three subjects: physics, chemistry, and biology. Examinees are required to select two of these subjects.

[Scope of Questions]

The scope of questions is as follows. What is taught in elementary and junior high schools in Japan is regarded as already learned and therefore to be included in the scope of the EJU. For each subject, the contents are grouped and presented by topic titles or key scientific terms.

Biology

The scope of questions will follow the scope of “Basic Biology” and “Advanced Biology” of the Course of Study for high schools in Japan.

I Evolution of organisms

1. Origin of life and evolution of cells
   (1) Origin of life
      Primitive Earth and chemical evolution
   (2) Evolution of cells
      First appearance of prokaryotic cells and emergence of photosynthetic organisms
      First appearance of eukaryotic cells and endosymbiosis
      Evolution of living organisms and changes in Earth’s environment
2. Genetic changes and changes in genetic composition
   (1) Genetic changes
      Mutations and changes in characteristics of organisms
      Nucleotide substitution, insertion and deletion
   (2) Changes in genetic composition
      Genes and chromosomes (gene loci, genotypes, and phenotypes)
      Sex chromosomes and autosomes
      Meiosis and fertilization
      Linkage and recombination
3. Mechanisms of evolution
   (1) Mechanisms of evolution
      Gene frequency and evolution
      Genetic drift and neutral theory
      Natural selection and adaptive evolution

(2) Speciation

Isolation and speciation

II Phylogeny and evolution of organisms

1. Phylogeny and evolution of organisms

(1) Phylogeny and classification of organisms

Nucleic acid sequences and amino acid sequences

Molecular evolution

Phylogenetic tree of vertebrates

Morphological and molecular phylogenetics

(2) Domains and phylogeny

Phylogeny of bacteria and archaea

Phylogeny of eukaryotes

(3) Human phylogeny and evolution

Evolution of primates

Evolution and migration of humans

III Life phenomena and substances

1. Cells and molecules

(1) Biological substances and cells

Substances that compose cells

Structure of biomembranes

Structure and function of cell organelles

Cytoskeleton

(2) Life phenomena and proteins

Structure and properties of proteins

Properties and functions of enzymes

Membrane transport proteins

Receptors and signal transduction

IV Metabolism

1. Metabolism and energy

(1) Anabolism and catabolism

(2) ATP and energy

2. Photosynthesis

(1) Photosynthesis and chloroplasts

Structure of chloroplast and photosynthetic pigments

Mechanism of photosynthesis and ATP synthesis

(2) Bacterial photosynthesis

3. Respiration and fermentation

(1) Respiration and mitochondria

Structure of mitochondria

Mechanism of respiration and ATP synthesis

(2) Fermentation and its mechanism
Lactic acid fermentation
Alcoholic fermentation

V Expression of genetic information and development

1. Genetic information and its expression

(1) Molecular structure and replication of DNA
Base complementarity and DNA replication
Cell cycle and DNA replication

(2) Genetic information and its expression
Transcription, splicing and translation
Transcription and translation in prokaryotes

2. Regulation of gene expression and development

(1) Regulation of gene expression
Regulation of transcription and regulation by gene regulators
Regulation of gene expression in prokaryotes
Regulation of gene expression in eukaryotes

(2) Development and gene expression
Gametogenesis, fertilization and cleavage in animals
Organizers and induction
Cell differentiation and morphogenesis
Expression of genes involved in animal development
Cell differentiation and regulation of gene expression

3. Techniques for genetic manipulation

(1) Techniques for genetic manipulation
Isolation and amplification of genes
Analysis of gene structure and expression

(2) Applications of gene technology
Application to food production, and application to medicine

VI Maintenance of the human internal environment

1. Regulation of the human body

(1) Bodily fluids and their circulation
Blood coagulation

(2) Autonomic nervous system and endocrine system

(3) Immunity

VII Animal responses and behavior

1. Animal responses

(1) Stimulus reception and response
Nervous system and neurons
Resting potentials and action potentials
Conduction of excitation and saltatory conduction

Synapses and signal transmission

Synaptic plasticity

(2) Receptor mechanisms

Stimulus reception and adequate stimulus

Transmission of information from receptors to the central nervous system

(3) Structure and function of the central nervous system

Structure and function of the brain

Structure and function of the spinal cord

Mechanism of reflex

(4) Mechanisms of effector organs

Structure of skeletal muscle

Conduction of excitation and muscle contraction

2. Animal behavior

(1) Innate behavior

(2) Acquired behavior and learning

VIII Plant growth and responses to the environment

1. Angiosperm reproduction and development

(1) Gametogenesis

(2) Double fertilization

(3) Embryogenesis

(4) Seed formation

2. Plant life and plant hormones

(1) Seed germination and responses to light

Seed dormancy and germination

Positive and negative photoblastic seeds

Photoreceptors

(2) Plant responses to the environment and growth

Functions of auxin

Responses to light and gravity

Tropisms and nastic movements

Stomatal opening and closure

(3) Flower bud formation and flower formation

Photoperiodism

Mechanism of flower bud formation

Vernalization

Flower structure and the ABC model

(4) Fruit development and maturation, defoliation, and fruit abscission

IX Ecology and environment

1. Population and community

(1) Population and its characteristics

Survivorship curve and age pyramid

Population density and intraspecific competition

Interaction within a population (sociality)

(2) Community and its characteristics

Predator and prey

Parasitism and symbiosis

Interspecific competition and ecological position (niche)

2. Ecosystems

(1) Floral succession and biomes

(2) Biomass production and matter cycles in ecosystems

Biomass production

Biomass movement and energy flow through trophic levels

Matter cycles (carbon cycle, nitrogen cycle, and nitrogen assimilation)

(3) Ecosystems and human activities

Ecosystems and biodiversity

Effects of human activities on ecosystems

Ecological balance and conservation